



Mid-term Examination

Time: 3 Hours

Total Marks: 20

Scenario No: 1 | Infrastructure Reconnaissance

(04 Marks)

Psychomotor Level 4: Complex Overt Response

Task: A private firm named **Whizminder** has commissioned you to conduct a **black-box Vulnerability Assessment and Penetration Testing (VAPT)** operation against its external infrastructure. As a cybersecurity expert, your task is to perform reconnaissance and open-source intelligence (OSINT) gathering activities without prior internal knowledge of the target environment. You are required to:

1. Perform an information gathering process to:
 - Identify the original domain of the organization.
 - Extract at least **two (2) unique subdomains** associated with the target.
 - Determine and document the associated IP address information of the identified domains/subdomains.
2. Configure and utilize appropriate reconnaissance tools or OSINT techniques to gather publicly available employee information that could potentially be used for social engineering attacks.

Scenario No: 2 | Service Enumeration & Vulnerability Scanning

(04 Marks)

Psychomotor Level 4: Complex Overt Response

Task: You are provided with a target IP address (TryHackMe room Blue) within the lab environment. Conduct a deep technical scan to identify potential entry points.

1. **Manipulate** Nmap flags to **execute** a comprehensive scan that identifies open ports, service versions, and the target's operating system.
2. **Operate** a nmap vulnerability scan to **detect** security flaws on the discovered services.

Scenario No: 3 | Exploitation Analysis & Mitigation

(10 Marks)

Psychomotor Level 3: Mechanism

Task: Based on the vulnerabilities identified in Scenario 2, demonstrate your ability to plan and exploit the identified vulnerabilities to gain remote (reverse shell) access to the target machine. (For Question 1, use the TryHackMe Blue Room environment. For Questions 2–5, use the Damn Vulnerable Web Application lab.)



Enhancing Digital Government Economy (EDGE) Digital Skills for Students



Host University: Chittagong University of Engineering and Technology

Host Dept./Institute/Center: Dept. of CSE, CUET

Venue: Southeast University, Batch Number: CADS-005

Training Track/Course Name: Applied Penetration Testing Engineer: Hands-on

Level: Advanced Level

1. Exploit the identified vulnerable service to obtain a Meterpreter reverse shell using the Metasploit Framework.
2. Exploit the XSS vulnerability to steal a user's cookie for account takeover.
3. Exploit the CSRF vulnerability to trick a user into changing their password and gain full control of the user's account.
4. Exploit the SQL injection vulnerability to dump all usernames and passwords from the database.
5. Exploit the Local File Inclusion (LFI) vulnerability to read sensitive information from the local machine, such as the contents of the passwd file.

Scenario No: 4 | Professional Reporting

(02 Marks)

Psychomotor Level 3: Mechanism

Task: Produce a professional summary of your lab activities.

1. Write a clear, professional report for the organization. Include a non-technical executive summary (2–3 sentences) for each task explaining the overall risk level to the organization's management.

Submission Requirements:

- Submit a PDF report containing screenshots for every command executed.
- Include the specific command syntax for all Nmap and hping3 tasks.
- Ensure the validation of findings is clearly documented.